Sample questions-Assignment 1

1. Define a signal and a system with two real-life examples of each.
2. Differentiate between continuous-time and discrete-time signals with suitable diagrams.
3. Classify $x(t) = e^{-2t}u(t)$ as even/odd/neither, and energy/power.
4. A Continuous Time signal is given by $x(t) = \cos\left(\frac{5\pi}{6}t\right)$. Is it periodic? If so, find its fundamental period.
5. Check whether $x[n] = \delta[n] + \delta[n-3]$ is causal and finite duration.
6. Determine the fundamental period of $x[n] = \sin\left(\frac{2\pi}{7}n\right)$.
7. Show that $x[n] = (0.8)^n u[n]$ is an energy signal. Compute its energy.
8. Consider the signal
$$x(t) = e^{-2t}u(t),$$
where $u(t)$ is the unit step function.
   (a) Determine whether $x(t)$ is an **energy signal** or a **power signal**.
   (b) Compute the corresponding value (energy or power).
9. Sketch $u(t)$, $\delta(t)$, and $r(t)$.
10. Express the rectangular pulse equal to 1 for $-2 \leq t < 3$, 0 elsewhere, using unit step functions.
11. Find and sketch $x(t) = u(t) - 2u(t-2) + u(t-4)$. Explain its shape.
12. Write the first five values of $x[n] = u[n] - u[n-5]$.
13. Construct $x[n] = \delta[n] + 2\delta[n-1] - \delta[n+2]$. Determine whether it is an energy or power signal and compute the value.
14. Determine whether
$$x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{\pi}{4}n\right)$$
is periodic. If yes, find the fundamental period.
15. If $x(t) = u(t) - u(t-4)$, sketch $x(t)$.
16. For $x(t) = e^{-t}u(t)$, sketch $y(t) = x(t-2)$.
17. Let $x(t) = u(t) - u(t-3)$. Sketch:
   (a) $y_1(t) = x(2t)$
   (b) $y_2(t) = x(-t+1)$
   (c) $y_3(t) = 3x(t) - x(t-1)$
18. Given $x[n] = u[n] - u[n-4]$, compute and sketch:
   (a) $x[n+2]$
   (b) $x[-n]$
19. If $x[n] = \{1, 2, 1, -1\}$ for $n = 0, 1, 2, 3$, compute
$$y[n] = 2x[n] - x[n-2].$$
Sketch both $x[n]$ and $y[n]$.